Misinformation Detection in News Headlines

Advanced NLP: Classification & Real-Time Fact Checking

Lachlan Johnston • Sebastian Litchfield • Jaiden Brocklebank
COMP6713 - Natural Language Processing | News Group

THE CHALLENGE OF MISINFORMATION

The Digital News Landscape

Modern news consumption has shifted to rapid, democratized digital platforms. While access is improved, it fosters **misinformation speed and scale**.

- → Sophisticated linguistic techniques to bypass skepticism.
- → False narratives gain traction before verification.
- → Requirement for **multi-modal detection systems**.

DUAL NLP PROBLEM ARCHITECTURE

Text Classification

Categorizing headlines into predefined labels (True, False, Mixed) using deep transformer architectures. Focuses on linguistic patterns and stylistic tendencies of journalism.

Information Retrieval

Expanding model capacity via LangChain to fact-check in real-time. Verifies claims against external world events using Google Serper and Gemini LLM agents.

DATASET SYNTHESIS & MAPPING

Combined Data Corpus (n=13,892)

We consolidated two major benchmarks to maintain high political domain consistency:

- **LIAR Dataset:** 12,836 short political statements from PolitiFact.
- **FakeNewsNet:** 1,056 samples focused on political fact-checks.
- **Refined Labels:** Simplified 6 original categories into 3: True (37%), Mixed (34%), False (29%).

13.8k

Total Annotated Samples

BASELINE: TF-IDF + LOGISTIC REG.

A statistical baseline to establish a performance floor. Configured with 10k features (unigrams & bigrams) with English stopwords removed.

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \log \frac{N}{\text{df}_t}$$

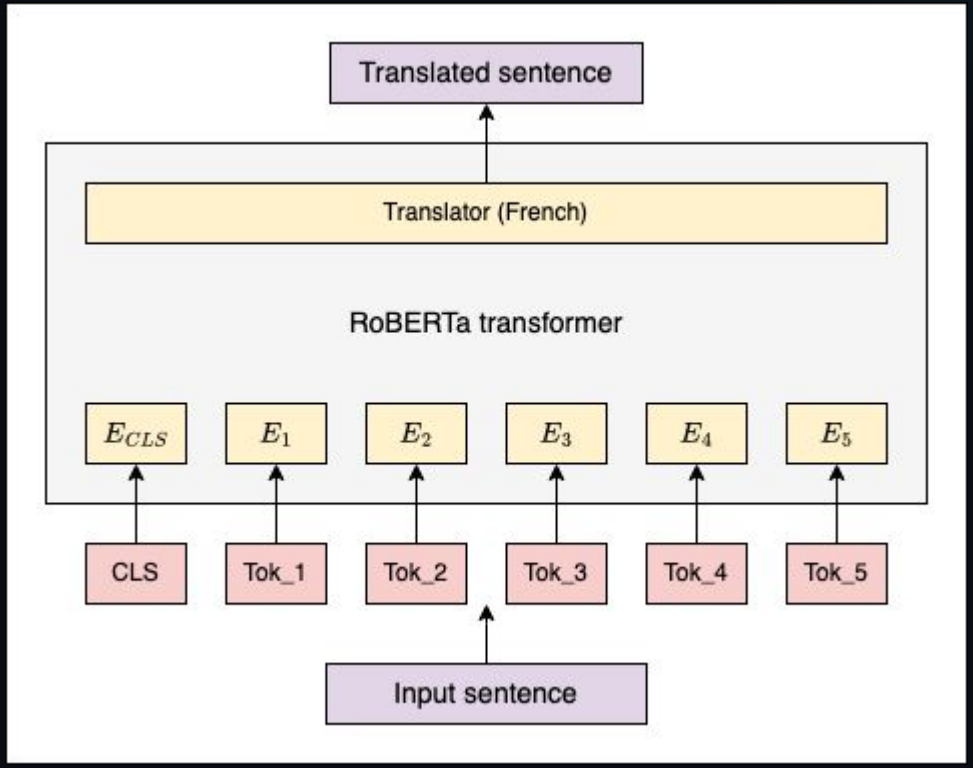
- → Balanced class weights for handling dataset skew.
- → Captures phrase-level signals like "Breaking news" or "Sources say".
- → Interpretability allows for clear error baseline analysis.

TRANSFORMER FINE-TUNING STRATEGY

BERT vs. RoBERTa

Deep training using 60/20/20 stratified splits.

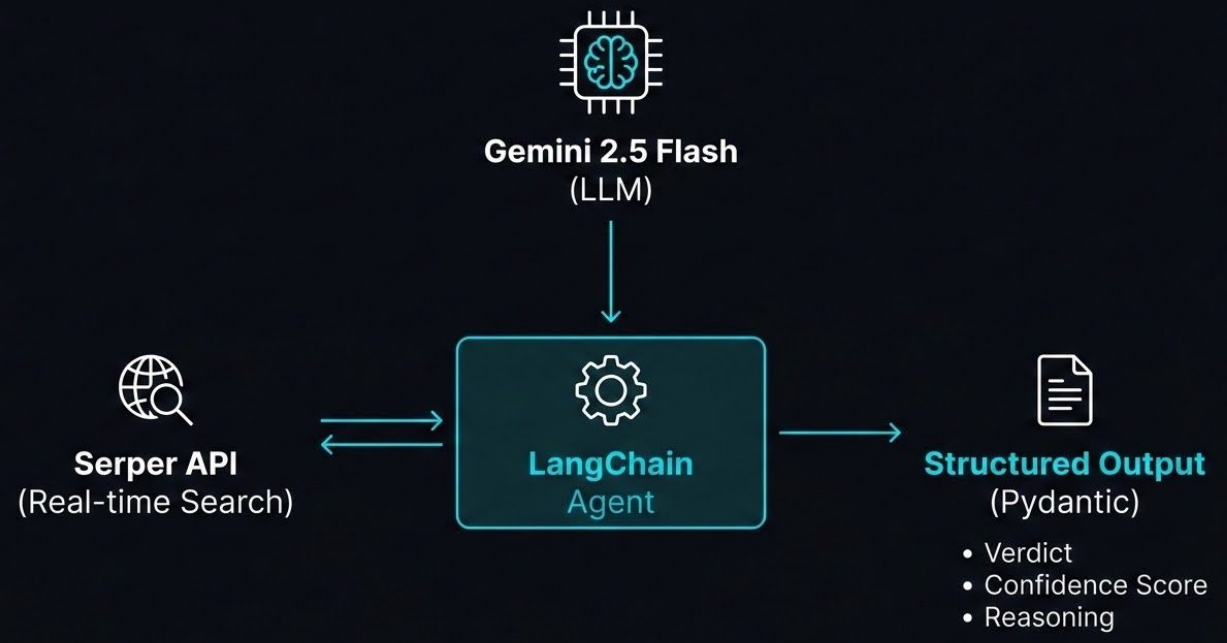
- **Fine-tuned BERT:** Learning rate 1e-5, AdamW optimizer, Linear warmup.
- **RoBERTa Enhancement:** Dynamic masking and removal of Next-Sentence Prediction (NSP).
- Early stopping enabled to prevent overfitting on short headlines.



LangChain Integrations

Integrated LangChain with Serper API to provide real-time fact checking functionalities.. **Gemini 2.5 Flash** serves as the reasoning agent.

- → Structured output using Pydantic.
- → Verdict + Confidence Score + Reasoning.
- → Weights reputable sources and evidence contradiction.



WORDNET LEXICAL EXPANSION

Synonym Swapping

Integrated WordNet to perform manual and automatic word swaps, expanding search query coverage for the IR system.

Concatenated Queries

Searches multiple query variants and concatenates results. Note: Higher search variance can lead to increased processing noise.

Efficiency Trade-off

Identified that while lexical expansion adds depth, modern LLMs like Gemini are often resilient to minor synonym changes.

SYSTEM PERFORMANCE COMPARISON

Model Architecture	Accuracy	Precision	Macro F1	Recall
Baseline (TF-IDF + LR)	0.4430	0.4414	0.4415	0.4425
Pre-trained BERT (Zero-shot)	0.3699	0.2005	0.1825	0.3326
Fine-tuned BERT	0.4933	0.5100	0.4810	0.4821
Fine-tuned RoBERTa	**0.5833**	0.5146	0.5772	0.5005
RoBERTa + LangChain + WordNet	**0.6000**	0.6054	0.5964	0.6000

Note: RoBERTa + LangChain provides the most balanced precision/recall metrics.

QUALITATIVE ERROR ANALYSIS

The "Temporal Truth" Problem

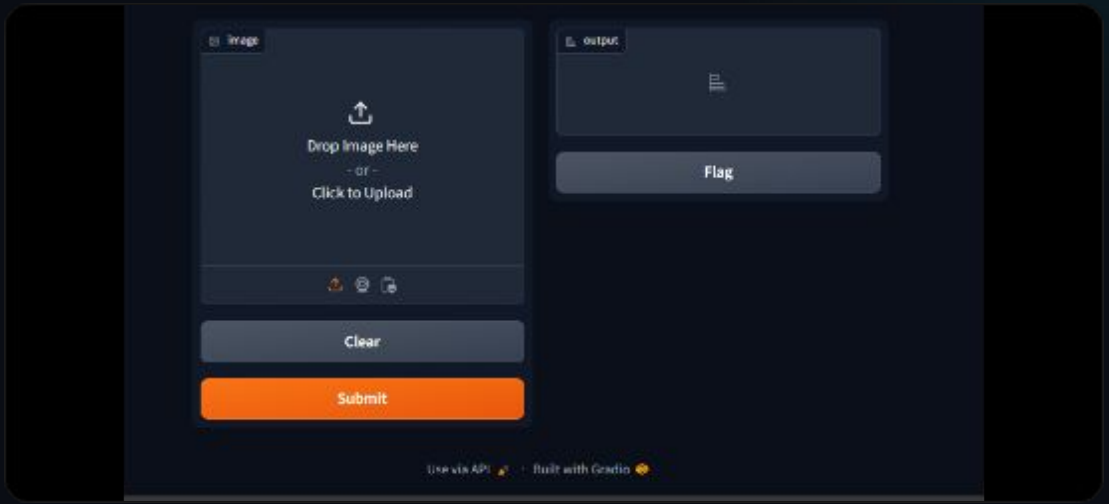
Many headlines were true at the time of dataset creation but have since changed.

- **Conflict:** Static Dataset Labels vs. Dynamic Search Evidence.
- **Observation:** Models struggle significantly with "False" labels due to nuanced misinformation tactics.
- **Solution:** Conservative integration logic where RAG only overrides RoBERTa if confidence thresholds are met.

RoBERTa Model	Predicted false	Predicted mixed	Predicted true
Actual false	9	6	5
Actual mixed	2	14	4
Actual true	4	4	12

SYSTEM INTERFACES & DEPLOYMENT

CLI Test: Rapid evaluation of headlines via command line. Supports --use-wordnet and model selection.



Gradio Demo: Interactive web UI for browser-based testing. Displays model predictions alongside RAG evidence side-by-side.

CONCLUSION & KEY FINDINGS

We successfully developed a layered misinformation detection suite. Moving from TF-IDF baselines to **RoBERTa architectures** provided significant accuracy gains. The integration of **LangChain RAG** addresses the limitation of static knowledge, creating a comprehensive verification tool for the modern digital news ecosystem.

- **RoBERTa** outperformed BERT in contextual understanding.
- **RAG** provides essential real-time verification.
- **WordNet** expansion offers depth but adds noise.
- **Temporal truth** remains a core challenge for NLP.

Questions?

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IMAGE SOURCES

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